Measure Theoretic Probability
Lesson 15
Random Variables and the σ-Field Generated by X

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Outline

1. Pulling Back Probabilities
2. The General Definition
3. Example: Measurability Depends on the σ-Field
4. The σ-Field Generated by X
5. Examples of $\sigma(X)$
6. Summary and Exercises

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The Central Question

Question

Let $X : \Omega \to \mathbb{R}$ and let $B \subseteq \mathbb{R}$. What is

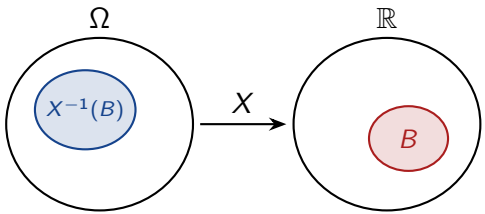
$$\mathbb{P}(X \in B) \, ?$$

Answer: pull back

The randomness lives in $(\Omega, \mathcal{F}, \mathbb{P})$, so we collect the outcomes that land in B:

$$X^{-1}(B) = \{\omega \in \Omega : \, X(\omega) \in B\}, \qquad \mathbb{P}(X \in B) = \mathbb{P}\big(X^{-1}(B)\big).$$

This only makes sense if $X^{-1}(B) \in \mathcal{F}$.



Notation

$X^{-1}(B)$ is the **inverse image** of B. It does **not** mean X is invertible as a function. It is the **set** of all ω with $X(\omega) \in B$, abbreviated $\{X \in B\}$.

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Borel Sets and the Definition of a Random Variable

Reminder: Borel σ-field on $\mathbb{R}$

$\mathcal{B} = \mathcal{B}(\mathbb{R})$ is the σ-field generated by the open intervals (a, b). It contains all open and closed intervals, half-open intervals, singletons $\{x\}$, and much more.

Definition (random variable)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A function $X : \Omega \to \mathbb{R}$ is a **random variable** (is $\mathcal{F}$-**measurable**) if

$$X^{-1}(B) \in \mathcal{F} \qquad \text{for every Borel set } B \in \mathcal{B}.$$

Remarks

- Measurability depends on **two** σ-fields: $\mathcal{F}$ on Ω and $\mathcal{B}$ on $\mathbb{R}$. We write nothing about $\mathcal{B}$ because it is the standard choice. With another σ-field $\mathcal{F}_2$ on $\mathbb{R}$ we would say "X is $\mathcal{F}/\mathcal{F}_2$-measurable".
- Not every subset of $\mathbb{R}$ is Borel (non-Borel sets exist). Contrary to a common belief, the Cantor set **is** Borel; it is a closed set.

Simple Random Variables

Recall

X is a simple random variable if it takes finitely many values and $\{X = x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. This mentions no Borel sets.

Proposition

For a function X with finitely many values $a_1, \ldots, a_k$, the following are equivalent:

1. $\{X = x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$;
2. $X^{-1}(B) \in \mathcal{F}$ for every Borel set B.

Proof

$(2 \Rightarrow 1)$: singletons are Borel and $\{X = x\} = X^{-1}(\{x\})$.
$(1 \Rightarrow 2)$: for Borel B, only the values a_i can contribute, so

$$X^{-1}(B) = \bigcup_{i:\ a_i \in B} \{X = a_i\},$$

a finite union of sets in $\mathcal{F}$, hence in $\mathcal{F}$. (If no a_i lies in B, the union is empty and $X^{-1}(B) = \varnothing \in \mathcal{F}$.) $\square$

Representation of Simple Random Variables

Claim

Every simple random variable can be written

$$X = \sum_{i=1}^{k} a_i \, I_{A_i}, \qquad a_i \in \mathbb{R} \text{ distinct}, \quad A_i \in \mathcal{F},$$

where $A_1, \ldots, A_k$ form a **partition** of Ω.

Proof

Let $a_1, \ldots, a_k$ be the distinct values of X and put $A_i = \{X = a_i\}$.

- $A_i \in \mathcal{F}$ by property (2) of the definition of simple.
- The A_i are pairwise disjoint because the a_i are distinct and X is a function.
- $\bigcup_i A_i = \Omega$ because X sends every ω to one of its values.

Hence $X(\omega) = a_i$ exactly when $\omega \in A_i$, which is the stated formula. □

Why "distinct"?

If $a_1 = a_3$, merge $A_1 \cup A_3$ into one set with a single coefficient. Any representation can be reduced to a distinct one.

Measurability Means Probabilities Are Determined

The point of measurability

X is $\mathcal{F}$-measurable precisely when every probability involving X is determined by the probabilities of sets in $\mathcal{F}$. This is why we insist on measurable pullbacks: if $X^{-1}(B) \notin \mathcal{F}$, then $\mathbb{P}(X \in B)$ is simply undefined.

Sub-σ-fields

If $\mathcal{G} \subseteq \mathcal{F}$ is a σ-field, X is $\mathcal{G}$-measurable if all inverse images $X^{-1}(B)$ lie in the smaller collection $\mathcal{G}$.

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Three Coin Flips: Two σ-Fields

Setup

$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$ and

$$X(\omega) = \begin{cases} 1, & \omega = HHH, \\ 0, & \text{otherwise.} \end{cases}$$

$\mathcal{F}_1 = \mathcal{P}(\Omega)$ (the power set)

Every subset of Ω is measurable. $\{X = 1\} = \{HHH\} \in \mathcal{F}_1$, $\{X = 0\} = \{HHH\}^c \in \mathcal{F}_1$, $\{X = 7\} = \varnothing \in \mathcal{F}_1$. So X is $\mathcal{F}_1$-measurable.

$\mathcal{F}_2 = \{\varnothing,\ \{HHT\},\ \{HHT\}^c,\ \Omega\}$

Here $\{X = 1\} = \{HHH\} \notin \mathcal{F}_2$, so $\mathbb{P}(X = 1)$ is not defined. X is **not** $\mathcal{F}_2$-measurable.

A question

$\mathcal{F}_1$ is large enough and $\mathcal{F}_2$ is too small. Is there a **smallest** σ-field for which X is measurable?

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Definition of $\sigma(X)$

Definition

$\sigma(X)$ is the **smallest** σ-field on Ω with respect to which X is measurable.

Construction

An intersection of σ-fields is a σ-field, so

$$\sigma(X) = \bigcap \{\mathcal{G} : \ \mathcal{G} \text{ is a } \sigma\text{-field on } \Omega \text{ and } X \text{ is } \mathcal{G}\text{-measurable}\}.$$

(The collection is non-empty since $\mathcal{P}(\Omega)$ belongs to it.)

In practice, for a simple X

Pull back the individual values $A_i = \{X = a_i\}$. These sets need not form a σ-field by themselves, so close them up under complements and countable unions. In other words,

$$\sigma(X) = \sigma(A_1, \ldots, A_k).$$

Theorem: What Sets Lie in $\sigma(X)$

Theorem

Let X be a simple random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.

(a) $\sigma(X)$ consists of all sets of the form

$$X^{-1}(H) = \{X \in H\}, \qquad H \subseteq \mathbb{R} \text{ arbitrary (not only Borel).}$$

(b) A simple random variable Y is $\sigma(X)$-measurable if and only if

$$Y = f(X) \qquad \text{for some function } f : \mathbb{R} \to \mathbb{R}.$$

Why it is useful

- (a) **characterises** the sets in $\sigma(X)$. To show $\sigma(X) \subseteq \mathcal{H}$ for another σ-field $\mathcal{H}$, take an element of $\sigma(X)$, write it as $X^{-1}(H)$, and continue.
- (b) says the information in X is exactly what lets you compute functions of X.

Plan

The proof is postponed: we will state it for **vectors** $(X_1, \ldots, X_n)$ and then recover the case $n = 1$.

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Example 1: An Indicator

Setup

$X = I_A$ with $A \in \mathcal{F}$ (so X takes the values 1 and 0). Take any $H \subseteq \mathbb{R}$ and look at $X^{-1}(H)$:

H **contains**	$X^{-1}(H)$
1 but not 0	A
0 but not 1	A^c
both 0 and 1	Ω
neither	$\varnothing$

Conclusion

$$\sigma(I_A) = \{\varnothing,\ A,\ A^c,\ \Omega\} = \sigma(A).$$

In the coin example, $\sigma(X) = \{\varnothing, \{HHH\}, \{HHH\}^c, \Omega\}$: it sits strictly between $\mathcal{F}_2$-type information and the power set.

Example 2: A Simple Random Variable with k Values

Setup

$X = \sum_{i=1}^{k} a_i I_{A_i}$ with distinct a_i and $A_1, \ldots, A_k$ a partition of Ω. For $H \subseteq \mathbb{R}$,

$$X^{-1}(H) = \bigcup_{i:\ a_i \in H} A_i = \bigcup_{i \in I} A_i, \qquad I = \{i :\ a_i \in H\} \subseteq \{1, \ldots, k\}.$$

Is the collection $\{\bigcup_{i \in I} A_i :\ I \subseteq \{1, \ldots, k\}\}$ already a σ-field?

- $\varnothing$: take $I = \varnothing$.
- Ω: take $I = \{1, \ldots, k\}$ (the A_i partition Ω).
- Complements: $\left(\bigcup_{i \in I} A_i\right)^c = \bigcup_{i \notin I} A_i$.
- Countable unions: only finitely many distinct unions exist, so these are finite unions of the same kind.

Conclusion

$\sigma(X) = \left\{\bigcup_{i \in I} A_i :\ I \subseteq \{1, \ldots, k\}\right\} = \sigma(A_1, \ldots, A_k)$. Each A_3, say, is recovered with $I = \{3\}$.

Example 3: A Constant Random Variable

Setup

$X(\omega) = c$ for all $\omega \in \Omega$. (Constants are random variables, "random with probability one".)

Computation

For $H \subseteq \mathbb{R}$:

$$X^{-1}(H) = \begin{cases} \Omega, & c \in H, \\ \varnothing, & c \notin H. \end{cases}$$

Conclusion

$$\sigma(X) = \{\varnothing, \Omega\}, \quad \text{the trivial } \sigma\text{-field.}$$

A constant carries no information, so it is measurable with respect to **every** σ-field.

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Summary

Key results

Concept / Result	Content
Inverse image	$X^{-1}(B) = \{\omega : X(\omega) \in B\}$, not an inverse function
Random variable	$X^{-1}(B) \in \mathcal{F}$ for every Borel B
Simple X	suffices to check $\{X = x\} \in \mathcal{F}$; $X = \sum a_i I_{A_i}$ over a partition
$\sigma(X)$	smallest σ-field making X measurable; $= \sigma(A_1, \ldots, A_k)$ if simple
Characterisation	$\sigma(X) = \{X^{-1}(H) : H \subseteq \mathbb{R}\}$; Y is $\sigma(X)$-measurable iff $Y = f(X)$
Examples	$\sigma(I_A) = \{\varnothing, A, A^c, \Omega\}$; constant $\Rightarrow \{\varnothing, \Omega\}$

Exercises

1. In the coin example, list every element of $\sigma(X)$ for $X = I_{\{HHH\}}$ and verify it is a σ-field contained in $\mathcal{P}(\Omega)$ but not contained in $\mathcal{F}_2$.
2. Let X be the number of heads in three flips. Find $\sigma(X)$ and count its elements.
3. Let $X = I_A + 2\,I_B$ where $A, B \in \mathcal{F}$ are disjoint. Write down the partition and describe $\sigma(X)$.
4. Show that if X is $\mathcal{G}$-measurable then $\sigma(X) \subseteq \mathcal{G}$.
5. Show that $\sigma(X) = \{\varnothing, \Omega\}$ if and only if X is constant (for a simple random variable).
6. Let $X = \sum a_i I_{A_i}$ be simple and $f : \mathbb{R} \to \mathbb{R}$ any function. Show that $f(X)$ is simple and $\sigma(X)$-measurable.

Outlook

Next lesson

Several random variables. The σ-field generated by a vector $(X_1, \ldots, X_n)$ and by a sequence $X_1, X_2, \ldots$ Then we prove the theorem on $\sigma(X)$ stated today by treating the vector case and setting $n = 1$.

Looking further ahead

With $\sigma(X_1), \sigma(X_2), \ldots$ in hand, independence of random variables and the tail σ-field of a sequence of random variables.